MIL-B-2679A
28 March 1963
SUPERSEDING
MIL-B-2679
23 August 1951

MILITARY SPECIFICATION

BRUSHES, WIRE, BOILER-TUBE, SECTIONAL TYPE, HAND PUSH

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy and the Air Force.

1. SCOPE

1.1 Scope. - This specification covers wire sectional type brushes for general purpose hand-push brushing of the interior walls of boiler tubes after attachment to rigid type push rods. Neither the complete brushes nor the sections should be used with motor-driven boiler tube cleaning outfits.

1.2 Classification. - Brushes shall be of but one class, that is, fixed-diameter, for sizes 2 to 4-inch-outside-diameter tubes, as specified (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-H-15424 - Hand Tools; Packaging of.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Material. - Material shall be of the best commercial quality and as specified herein.

3.2 Construction. - All sectional wire boiler-tube brushes shall be made up of tubular sections; each section shall consist of steel wire fagots mounted on a tubular steel or brass drum. The drums or sections shall be mounted on a steel spindle having screwed end fittings designed to clamp sections securely together.

3.3 Fagots. - Each fagot shall consist of approximately 40 hardened and polished steel wires, 0.009 inch in diameter. The fagots shall be doubled and securely fastened at the middle, through a hole in the

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drum, by a steel or bronze binding wire, or by stapling. The ends of each fagot shall be cut evenly and at such a point as to produce a brush of the outside diameter required. The fagots shall be arranged in 4 equidistant rows around the circumference of each drum; the rows shall be staggered so that the surface is uniformly covered.

3.4 Bearing collars. - Each drum shall have a bearing collar, fitted loosely over the spindle, to prevent wear on the binding wires.

3.5 End fittings. - The screwed end fittings shall be machined from hexagon steel or brass rod, with sufficient length left unturned to permit gripping with a wrench. The end clamp collars or washers for brushes shall be conical or flat in shape and made of steel or brass. The outer end fittings shall be rounded at the end of the hexagonal section. The inner end fitting shall be tapped 3/8 inch I.P.S., at the end of the hexagonal section in accordance with the requirements of Handbook H-28. The fittings shall be mounted on spindles, threaded on each end with at least UNC-2A fit, of sufficient length to permit clamping the sections of the brush together. The spindles shall conform to the sizes shown in table I. The assembled brush shall be similar to figure 1.

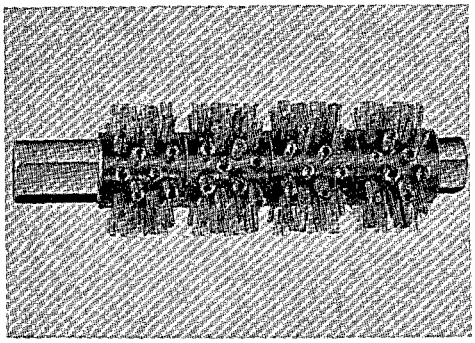


Figure 1 - Assembled boiler tube brush.

Table I. Dimensions

Size (outside diameter) tube	Thickness of wall	Diameter over fagots 1/	Outside diameter of drum	Width of drum	Inside Diameter of collar (max.)	Thickness of collar	Size of spindle (max.)	Number of fagots per section
Inches	Mils	Inches	Inches	Inches	Inch	Inch	Inches	
2	95	1 27/32	1	1 1/8	1/2	1/16	3/8 by 6	40
2	134	1 49/64	1	1 1/8	1/2	1/16	3/8 by 6	40
2	165	1 45/64	1	1 1/8	1/2	1/16	3/8 by 6	40
2 1/2	109	2 5/16	1 3/8	1 1/8	9/16	1/16	7/16 by 6 1/4	60
3	109	2 13/16	1 3/4	1 1/4	5/8	1/4	1/2 by 5 1/2	80
3	120	2 51/64	1 3/4	1 1/4	5/8	1/4	1/2 by 5 1/2	80
3	165	2 45/64	1 3/4	1 1/4	5/8	1/4	1/2 by 5 1/2	80
3 1/2	120	3 19/64	2 1/8	1 1/4	11/16	1/4	9/16 by 7 3/4	100
4	134	3 49/64	2 1/2	1 1/4	3/4	1/4	5/8 by 7 3/4	120
4	165	3 45/64	2 1/2	1 1/4	3/4	1/4	5/8 by 7 3/4	120
4	203	3 5/8	2 1/2	1 1/4	3/4	1/4	5/8 by 7 3/4	120

1/ A minus tolerance of 1/32 inch in diameter over fagots will be permitted.

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3.6 Sections per brush. - Two-and 2 1/2-inch sectional brushes shall have four sections; 3, 3 1/2 and 4-inch brushes shall have three sections.

3.7 Dimensions. - Dimensions of all brushes and brush sections shall be as shown in table I.

3.8 Brush sections (refills). - When ordered separately, brush sections shall conform to the requirements of 3.3 and 3.7. Brush section (refills) shall be similar to figure 2.

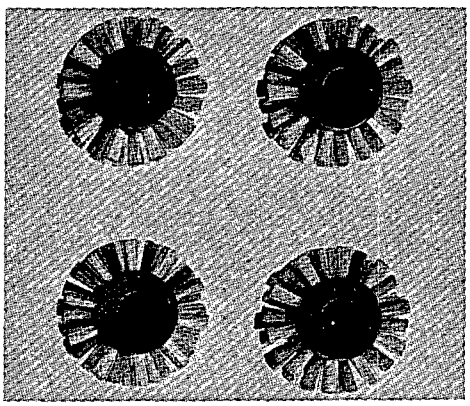


Figure 2 - Brush sections (refills).

3.9 Marking. - The size of tube (outside diameter and thickness of wall) shall be stamped on each brush section.

3.10 Workmanship. - Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the perfromance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification: where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Lot. - For purposes of sampling, a lot shall consist of all brushes or refill sets of one class and size offered for delivery at one time.

4.2.2 Sampling for visual examination. - A random sample of brushes or refill sets shall be selected from each lot in accordance with Standard MIL-STD-105 at inspection level II, and Acceptable Quality Level (AQL) equals 1.5 percent defective. Each of the sample units shall be subjected to the visual examination specified in 4.3.1.

4.2.3 Sampling for dimensional examination. - A random sample of brushes or refill sets shall be selected from each lot in accordance with Standard MIL-STD-105, at inspection level L-7 and Acceptable Quality Level equals 2.5 percent defective. Each of the sample units shall be subjected to the dimensional examination specified in 4.3.1.1.

4.2.4 Sampling for bristle and fagot count. - Random samples of brushes or refill sets shall be selected from each lot in accordance with Standard MIL-STD-105, at inspection level L-4 and shall be subjected to the examination specified in 4.3.1.2. The acceptable quality level shall be equal to 1.5 percent defective.

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4.3 Inspection. -

4.3.1 Visual examination. - Each of the sample brushes or refill sets selected in accordance with 4.2.2 shall be subjected to visual examination (other than dimensional examination and bristle and fagot count) to determine compliance with this specification. If the number of nonconforming brushes exceeds the acceptance number shown in 4.2.2, the entire lot shall be rejected.

4.3.1.1 Dimensional examination. - Each of the samples selected in accordance with 4.2.3 shall be examined for dimensions specified in applicable paragraphs and tables to determine compliance with the requirements of this specification. If the number of nonconforming items exceeds the acceptance number specified in 4.2.3, the entire lot shall be rejected.

4.3.1.2 Bristle and fagot count. - The number of wire ends shall be carefully counted in one fagot of each of the sample brushes selected in accordance with 4.2.4. The fagot to be counted shall be selected from a different row in each of the sample brushes or refill sets. Counting shall be continued by counting of rows per section and of fagots per section to determine conformance with number requirements of 3.3 and table I. If the number of nonconforming items exceeds the acceptance level shown in 4.2.4, the entire lot shall be rejected.

4.4 Inspection of preparation for delivery. - Sample brushes and packs shall be selected and inspected in accordance with Specification MIL-H-15424 to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. -

5.1.1 Brushes and brush sections (refills) shall be preserved and packaged level A or C, as specified (see 6.1), packed level A, B or C as specified (see 6.1) and marked in accordance with Specification MIL-H-15424. Brushes shall be packaged by units, section refills by sets sufficient to renew one brush.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of brushes (see 1.2 and table I).
(c) Level of packaging and level of packing required (see 5.1).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. *RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.*

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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